

Iwasawa Theory: The p -adic Machine for BSD and Beyond

Diego Rincón

Independent

Abstract

Iwasawa theory is the p -adic algebraic machine that drives the known cases of the Birch–Swinnerton-Dyer conjecture. The central idea is to study arithmetic objects not over $\mathbb{Q}$ but over the full cyclotomic $\mathbb{Z}_p$ -extension $\mathbb{Q}_\infty = \bigcup_n \mathbb{Q}(\zeta_{p^n})$, where the Galois group $\Gamma = \text{Gal}(\mathbb{Q}_\infty/\mathbb{Q}) \cong \mathbb{Z}_p$ admits a completed group ring $\Lambda = \mathbb{Z}_p[[\Gamma]]$. The Iwasawa main conjecture (proved for elliptic curves over $\mathbb{Q}$ by Skinner–Urban 2014 [2]) identifies the characteristic ideal of the Pontryagin dual of the Selmer group with the p -adic L -function: $\text{char}_\Lambda(\text{Sel}_p(E/\mathbb{Q}_\infty)^\vee) = (L_p(E))$ in Λ . We build the full machine: the Iwasawa algebra, the Selmer group over the tower, the p -adic L -function, the main conjecture, and the descent from the main conjecture to BSD. The key input is Kolyvagin’s Euler system of Heegner points, which bounds the Selmer group from above and forces $\text{rank } E(\mathbb{Q}) \leq \text{analytic rank}$. We also locate the three open ingredients needed to push this machine to higher rank and to the arithmetic GGP conjecture of [7].

1 The Iwasawa Algebra

Definition 1 (Cyclotomic tower). Fix a prime p and a compatible system of primitive p^n -th roots of unity ζ_{p^n} (so $\zeta_{p^{n+1}}^p = \zeta_{p^n}$). The *cyclotomic $\mathbb{Z}_p$ -extension* is

$$\mathbb{Q}_\infty = \bigcup_{n \geq 0} \mathbb{Q}_n, \quad \mathbb{Q}_n = \mathbb{Q}(\zeta_{p^n}),$$

with $\Gamma_n = \text{Gal}(\mathbb{Q}_n/\mathbb{Q}) \cong (\mathbb{Z}/p^n)^\times$ and $\Gamma = \text{Gal}(\mathbb{Q}_\infty/\mathbb{Q}) = \varprojlim \Gamma_n \cong \mathbb{Z}_p$.

Definition 2 (Iwasawa algebra). The *Iwasawa algebra* is the completed group ring

$$\Lambda = \mathbb{Z}_p[[\Gamma]] = \varprojlim_n \mathbb{Z}_p[\Gamma/\Gamma^{p^n}].$$

Choosing a topological generator $\gamma \in \Gamma$ gives an isomorphism $\Lambda \cong \mathbb{Z}_p[[T]]$ via $\gamma \mapsto 1 + T$. Thus Λ is a local, Noetherian, regular ring of dimension 2 — in particular a UFD.

Proposition 3 (Structure theorem for Λ -modules). *Every finitely generated torsion Λ -module M is pseudo-isomorphic (isomorphism up to finite kernel and cokernel) to*

$$\bigoplus_i \Lambda/(f_i(T)^{e_i}) \oplus \bigoplus_j \Lambda/(p^{n_j}),$$

where $f_i \in \mathbb{Z}_p[T]$ are distinguished polynomials (leading coefficient 1, other coefficients in $p\mathbb{Z}_p$). The characteristic ideal

$$\text{char}_\Lambda(M) = \left(\prod_i f_i(T)^{e_i} \cdot p^{\sum_j n_j} \right) \subset \Lambda$$

is well-defined (independent of the pseudo-isomorphism) and is the central invariant of Iwasawa theory.

2 Selmer Groups over the Tower

Definition 4 (Selmer group). Let $E/\mathbb{Q}$ be an elliptic curve, p an odd prime of good ordinary reduction for E . The p -adic Selmer group over a field F is

$$\text{Sel}_p(E/F) = \ker \left(H^1(G_F, E[p^\infty]) \longrightarrow \prod_v \frac{H^1(G_{F_v}, E[p^\infty])}{E(F_v) \otimes \mathbb{Q}_p/\mathbb{Z}_p} \right),$$

where the product is over all places v of F and the local condition at v is the image of the Kummer map.

Remark 5 (The Selmer group over the tower). Over $\mathbb{Q}_\infty$, the Selmer group $\text{Sel}_p(E/\mathbb{Q}_\infty) = \varinjlim_n \text{Sel}_p(E/\mathbb{Q}_n)$ is a discrete Λ -module. Its Pontryagin dual $X_\infty = \text{Sel}_p(E/\mathbb{Q}_\infty)^\vee$, i.e., $\text{Hom}_{\mathbb{Z}_p}(\text{Sel}_p(E/\mathbb{Q}_\infty), \mathbb{Q}_p/\mathbb{Z}_p)$, is a finitely generated torsion Λ -module (Kato 2004 [1], building on Rohrlich's non-vanishing of twists).

Theorem 6 (Kato's Euler system bound). *For $E/\mathbb{Q}$ with p odd and good ordinary, Kato [1] constructs an Euler system $\{c_n\}$ for $T_p(E)$ using Beilinson elements in $K_2(\mathbb{Q}(\zeta_n))$. The Euler system machinery (Kolyvagin, Rubin) gives*

$$\text{char}_\Lambda(X_\infty) \mid (L_p(E)) \quad \text{in } \Lambda.$$

That is, the characteristic ideal of the Selmer group divides the p -adic L -function.

3 The p -adic L -function

Theorem 7 (Mazur–Swinnerton-Dyer, Vishik, Amice–Vélu). *For $E/\mathbb{Q}$ with good ordinary reduction at p , there exists a p -adic L -function $L_p(E, \cdot) : \text{Hom}_{\text{cts}}(\Gamma, \mathbb{Z}_p^\times) \rightarrow \mathbb{Z}_p$ characterized by the interpolation property: for every finite-order character $\chi : \Gamma \rightarrow \mu_{p^n}$,*

$$L_p(E, \chi) = \frac{e_p(\chi)}{\Omega_E^\pm} \cdot L(E \otimes \chi, 1),$$

where $e_p(\chi)$ is an explicit Euler factor at p , $\Omega_E^\pm$ are the real and imaginary periods of E , and the sign is determined by the sign of $\chi(-1)$. Via $\Lambda \cong \mathbb{Z}_p[[T]]$, the p -adic L -function corresponds to a power series $f_E(T) \in \Lambda$.

Remark 8 (Order of vanishing and BSD). Setting $\chi = \mathbf{1}$ (the trivial character) and expanding near $T = 0$:

$$\text{ord}_{T=0} f_E(T) = \text{ord}_{s=1} L(E, s)$$

is predicted by BSD. The p -adic L -function is the bridge between the complex L -function and the algebraic Selmer group.

4 The Iwasawa Main Conjecture

Conjecture 9 (Iwasawa main conjecture for elliptic curves). *For $E/\mathbb{Q}$ with good ordinary reduction at p :*

$$\text{char}_\Lambda(X_\infty) = (f_E(T)) \quad \text{in } \Lambda.$$

The two inclusions have different proofs and depths.

Theorem 10 (Skinner–Urban 2014 [2]). *Under mild hypotheses:*

1. $p \geq 5$ is odd,
2. E has good ordinary reduction at p ,
3. $\bar{\rho}_{E,p}$ is irreducible and p -distinguished,
4. the sign in the functional equation of $L(E, s)$ is -1 (rank 1 case),

the divisibility $(f_E(T)) \mid \text{char}_\Lambda(X_\infty)$ holds. Together with Kato’s bound (Theorem 6), the main conjecture holds:

$$\text{char}_\Lambda(X_\infty) = (f_E(T)).$$

Remark 11 (The method: p -adic Eisenstein congruences). Skinner–Urban construct a 3-variable p -adic family of automorphic forms on $\text{GSp}(4)$ by Eisenstein series whose specializations congruent to the Klingen Eisenstein lift of E . The p -adic L -function appears in the constant term; the Selmer group appears via the Bloch–Kato exponential map and Galois cohomology. The equality of ideals in Λ follows from the uniqueness of the characteristic ideal.

5 Descent: From the Main Conjecture to BSD

Theorem 12 (Main conjecture $\Rightarrow$ BSD (rank and finiteness)). *Assume the Iwasawa main conjecture (Conjecture 9) and that $\text{III}(E/\mathbb{Q})[p^\infty]$ is finite. Then:*

1. $\text{ord}_{s=1} L(E, s) = \text{rank } E(\mathbb{Q})$ (rank part of BSD),

2. $\#III(E/\mathbb{Q}) \cdot (\text{explicit product of local terms}) = |L^*(E, 1)/\Omega_E^+|_p^{-1}$ (leading coefficient up to p -part).

Proof sketch. Specializing the characteristic ideal identity at $T = 0$:

$$\text{char}_\Lambda(X_\infty)|_{T=0} = f_E(0) = L_p(E, \mathbf{1}).$$

By the control theorem (Mazur), $X_\infty|_{T=0}$ is pseudo-isomorphic to the p -part of the Selmer group over $\mathbb{Q}$: $\text{Sel}_p(E/\mathbb{Q})^\vee$. The p -adic Birch–Swinnerton-Dyer formula then reads

$$\text{ord}_{T=0} f_E = \text{rank } E(\mathbb{Q}),$$

and $f_E(0) = L_p(E, \mathbf{1})$ encodes the leading term of $L(E, s)$ at $s = 1$ via the interpolation property. The finiteness of $III[p^\infty]$ converts the Selmer exact sequence $0 \rightarrow E(\mathbb{Q}) \otimes \mathbb{Q}_p/\mathbb{Z}_p \rightarrow \text{Sel}_p \rightarrow III[p^\infty] \rightarrow 0$ into a formula for $\#III[p^\infty]$ in terms of $f_E(0)$. $\square$

Corollary 13 (BSD for analytic rank ≤ 1). *For elliptic curves $E/\mathbb{Q}$ with $\text{ord}_{s=1} L(E, s) \leq 1$, combining Skinner–Urban (Theorem 10) with Kolyvagin’s Heegner point Euler system:*

1. Rank 0: $L(E, 1) \neq 0$ implies $E(\mathbb{Q})$ finite and $III(E/\mathbb{Q})$ finite.
2. Rank 1: $L'(E, 1) \neq 0$ implies $\text{rank } E(\mathbb{Q}) = 1$ and $III(E/\mathbb{Q})$ finite.

6 Kolyvagin’s Euler System

Definition 14 (Heegner points). Let K be an imaginary quadratic field satisfying the Heegner hypothesis: every prime $\ell \mid N$ splits in K . For each n , the ideal $\mathfrak{a}_n = (\mathbb{Z} + p^n \mathcal{O}_K)$ defines a CM point $x_n \in X_0(N)(\overline{\mathbb{Q}})$ with complex multiplication by $\mathcal{O}_{K, p^n}$. The image $y_n = \phi(x_n) \in E(\overline{\mathbb{Q}})$ via the modular parametrization $\phi : X_0(N) \rightarrow E$ is the *Heegner point* at level n .

Theorem 15 (Gross–Zagier, Kolyvagin). 1. (Gross–Zagier 1986 [3]) *If $\text{ord}_{s=1} L(E, s) = 1$, then the trace $y_K = \text{Tr}_{K_1/K}(y_1) \in E(K)$ is non-torsion, and*

$$L'(E/K, 1) = \frac{2\|\omega\|^2}{\sqrt{|d_K|} |N(E, K)|} \cdot \hat{h}(y_K),$$

the Néron–Tate height of the Heegner point.

2. (Kolyvagin 1988 [4]) *If y_K is non-torsion, then $\text{rank } E(\mathbb{Q}) = 1$ and $III(E/\mathbb{Q})$ is finite. The system $\{y_n\}$ is an Euler system for $T_p(E)$: the norm-compatible collection satisfies $\text{Tr}_{K_{n+1}/K_n}(y_{n+1}) = a_p(E) \cdot y_n - y_{n-1}$, and Kolyvagin’s derivative operators produce annihilators of $\text{Sel}_p(E/\mathbb{Q})$ from the Heegner points.*

7 Open Ingredients for Higher Rank

The Iwasawa machine as described works completely for analytic rank ≤ 1 . For rank ≥ 2 , three ingredients are missing (see also [7]):

1. **Euler system in rank ≥ 2 :** Kolyvagin’s Heegner points live on $E(K)$ and give one point (rank 1). For rank r , one needs an Euler system built from r -dimensional cycles — conjecturally the *Beilinson–Flach classes* on E^r or Selmer classes from r -fold Kuga–Sato varieties. No such system is constructed in full generality.
2. **p -adic L -function for $\mathrm{GL}(2)^r$:** The Mazur–Swinnerton-Dyer p -adic L -function interpolates $L(E \otimes \chi, 1)$ for characters χ of Γ . For the arithmetic GGP conjecture (see [7]), one needs a p -adic L -function interpolating the triple, quadruple, ..., product L -functions $L(E_1 \otimes \cdots \otimes E_r, s)$ at the central value. The Darmon–Harris–Rotger program constructs these for $r = 3$; the general case is open.
3. **Explicit reciprocity law:** The bridge between the p -adic L -function and the Euler system classes is an *explicit reciprocity law*: a formula $\langle \exp^*(c), \log(\sigma) \rangle = L_p(\cdot)$, where $\exp^*$ is the Bloch–Kato dual exponential and σ is a Selmer class. For rank 1, this is Kato’s explicit reciprocity law (a p -adic refinement of the Birch–Swinnerton-Dyer regulator). For rank ≥ 2 , the corresponding formula requires the p -adic height pairing on the full Selmer group, and the height is not yet shown to be non-degenerate in general.

Remark 16 (Bloch–Kato as the target). The Bloch–Kato conjecture [6] generalizes BSD to arbitrary motives M : for a pure motive M of weight -1 ,

$$\mathrm{ord}_{s=0} L(M, s) = \dim_{\mathbb{Q}_p} H_f^1(G_{\mathbb{Q}}, V) \quad \text{and} \quad L^*(M, 0) \sim \frac{\#H_f^1/H_f^1(\mathrm{int})}{\#H^0 \cdot \#H^2} \cdot \det(\langle \cdot, \cdot \rangle_{\mathrm{BK}}),$$

where H_f^1 is the Bloch–Kato Selmer group and $\langle \cdot, \cdot \rangle_{\mathrm{BK}}$ is the Bloch–Kato height pairing. BSD is the case $M = h^1(E)(1)$. The Iwasawa main conjecture for a general motive M is: $\mathrm{char}_{\Lambda}(H_f^1(G_{\mathbb{Q}_{\infty}}, V)^{\vee}) = (L_p(M))$. This is the arithmetic frontier — proved for $\mathrm{GL}(1)$ (Kummer theory, Herbrand–Ribet) and for $\mathrm{GL}(2)$ elliptic curves (Skinner–Urban), conjectural in general.

8 The Machine: Summary

Theorem 17 (Iwasawa machine). *The following chain holds unconditionally for elliptic curves $E/\mathbb{Q}$ with p good ordinary and $\bar{\rho}_{E,p}$ irreducible:*

<i>Step</i>	<i>Statement</i>	<i>Reference</i>
1	$L_p(E) \in \Lambda$ interpolates $L(E \otimes \chi, 1)$	[5]
2	$X_\infty = \text{Sel}_p(E/\mathbb{Q}_\infty)^\vee$ is fin. gen. torsion	[1]
3	$\text{char}_\Lambda(X_\infty) \mid (L_p(E))$	[1]
4	$(L_p(E)) \mid \text{char}_\Lambda(X_\infty)$	[2]
5	Main conjecture: $\text{char} = (L_p)$	3 + 4
6	Specialize at $T = 0$: rank and III bound	Control thm
7	Heegner point non-torsion (rank 1 sign)	[3, 4]
8	BSD for $\text{ord}_{s=1} L(E, s) \leq 1$	5 + 7

Steps 3–5 are the arithmetic core; step 7 provides the geometric input; the gap at rank ≥ 2 is the missing Euler system (step 7 generalized).

Remark 18 (Position in the body). The Iwasawa main conjecture is the p -adic shadow of the full BSD formula (Proposition 12). BSD itself is a special case of Beilinson’s conjectures (see [8]), which are in turn the motivic upgrade of the Tate conjecture (see [9]). The arithmetic GGP conjecture (see [7]) encodes the higher-rank BSD formula via periods of unitary Shimura varieties; all three open ingredients above are the obstacles to completing that program.

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